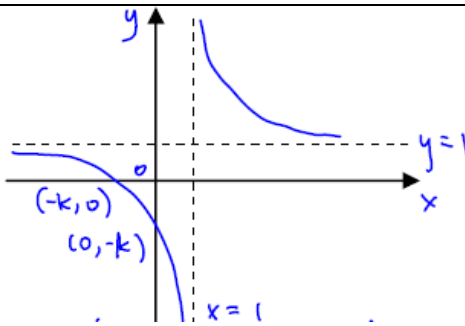


No.	Solutions	Comments
1	Let \$x\$, \$y\$ and \$z\$ be the cost of a ticket for “under 16 years”, “between 16 and 65 years”, and “over 65 years” categories respectively. $9x + 6y + 4z = 162.03 \quad \text{-----}(1)$ $7x + 5y + 3z = 128.36 \quad \text{-----}(2)$ $10x + 4y + 5z = 158.50 \quad \text{-----}(3)$ $\Rightarrow x = 7.65, \quad y = 9.85, \quad z = 8.52$ Cost of a ticket for the “under 16 years”, “between 16 and 65 years”, and “over 65 years” categories are \$7.65, \$9.85 and \$8.52 respectively.	Do take note of the context of the question, we are solving for cost, hence answers should be left correct to 2 decimal places.
2i)	$\int \frac{x^3}{1+x^4} dx = \frac{1}{4} \int \frac{4x^3}{1+x^4} dx$ $= \frac{1}{4} \ln(1+x^4) + C$	(i) Do you know why answer is given as $\frac{1}{4} \ln(1+x^4)$ instead of $\frac{1}{4} \ln 1+x^4 $?
2ii)	$u = x^2 \Rightarrow \frac{du}{dx} = 2x$ $\int \frac{x}{1+x^4} dx = \frac{1}{2} \int \frac{2x}{1+x^4} dx$ $= \frac{1}{2} \int \frac{1}{1+u^2} du$ $= \frac{1}{2} \tan^{-1} u + C$ $= \frac{1}{2} \tan^{-1}(x^2) + C$	(ii) Do remember to leave final answer in terms of x . Do remember to put in constant of integration for both parts (i) & (ii).
2iii)	$\int_0^1 \left(\frac{x}{1+x^4} \right)^2 dx = 0.186 \quad (\text{correct to 3 dp})$	(iii) GC can be used due to key words in the question such as “evaluate” and “correct to 3 decimal places”.
3i)	$u_2 = \frac{3(2)-1}{6} = \frac{5}{6}$ $u_3 = \frac{3\left(\frac{5}{6}\right)-1}{6} = \frac{1}{4}$	Not in H2 Math 9758 syllabus Exact value is required.
3ii)	$l = \frac{3l-1}{6}$ $\Rightarrow 6l - 3l = -1$ $\Rightarrow l = -\frac{1}{3}$	Not in H2 Math 9758 syllabus (ii) Read the question carefully, GC cannot be used to find l since working needs to be shown & exact answer is required.
3iii)	Let P_n be the statement $u_n = \frac{14}{3} \left(\frac{1}{2} \right)^n - \frac{1}{3}, \quad n \in \mathbb{Z}^+.$ LHS of $P_1 = u_1 = 2$ (given)	Not in H2 Math 9758 syllabus (iii) It is important to be precise in your written work. <u>Bad Presentation</u> When $n = 1$,

	RHS of $P_1 = \frac{14}{3} \left(\frac{1}{2} \right)^1 - \frac{1}{3} = \frac{6}{3} = 2 = \text{LHS}$ $\therefore P_1$ is true. Assume P_k is true for some $k \geq 1$. i.e. $u_k = \frac{14}{3} \left(\frac{1}{2} \right)^k - \frac{1}{3}$ We want to show P_{k+1} is true, i.e. $u_{k+1} = \frac{14}{3} \left(\frac{1}{2} \right)^{k+1} - \frac{1}{3}$ LHS of $P_{k+1} = u_{k+1}$ $= \frac{3u_k - 1}{6}$ $= \left(\frac{1}{2} \right) \left[\frac{14}{3} \left(\frac{1}{2} \right)^k - \frac{1}{3} \right] - \frac{1}{6}$ $= \frac{14}{3} \left(\frac{1}{2} \right)^{k+1} - \frac{1}{6} - \frac{1}{6}$ $= \frac{14}{3} \left(\frac{1}{2} \right)^{k+1} - \frac{1}{3} = \text{RHS of } P_{k+1}$ $\therefore P_{k+1}$ is true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true $\therefore u_n = \frac{14}{3} \left(\frac{1}{2} \right)^n - \frac{1}{3}$ by induction for all $n \in \mathbb{Z}^+$.	LHS = 2 = RHS Do ensure that your conclusion is written precisely and the essence of the induction process needs to be seen in your conclusion, i.e. ✓ $P_k \text{ true} \Rightarrow P_{k+1} \text{ true}$ Hence the following conclusion ✗ $P_k \text{ true}$ and so is P_{k+1} is <u>not acceptable</u>
4i	$\frac{AC}{\sin \frac{3\pi}{4}} = \frac{AB}{\sin \left(\pi - \frac{3\pi}{4} - \theta \right)}$ $AC = \frac{\sqrt{2}}{2} \left(\frac{1}{\sin \frac{\pi}{4} \cos \theta - \cos \frac{\pi}{4} \sin \theta} \right)$ $= \frac{\sqrt{2}}{2 \left(\frac{\sqrt{2}}{2} \cos \theta - \frac{\sqrt{2}}{2} \sin \theta \right)}$ $= \frac{1}{\cos \theta - \sin \theta}$	This question proved to be difficult for a lot of students as they were unable to find AC. (i) Do note that Trigonometry is an assumed knowledge from Additional Mathematics.
4ii	Given θ is sufficiently small,	(ii) Take note of “hints” given in the question • Small angle approximation • Need to get a series $\rightarrow$ need to apply series expansion Learn when and how to apply binomial expansion.

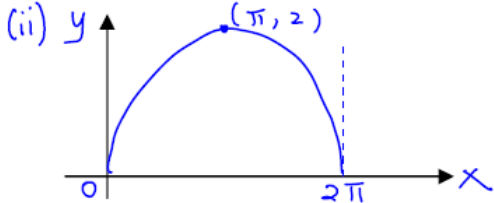
	$AC \approx \frac{1}{1 - \frac{\theta^2}{2} - \theta}$ $= \left[1 + \left(-\theta - \frac{\theta^2}{2} \right) \right]^{-1}$ $\approx 1 + (-1) \left(-\theta - \frac{\theta^2}{2} \right) + \frac{(-1)(-2)}{2!} \left(-\theta - \frac{\theta^2}{2} \right)^2$ $\approx 1 + \theta + \frac{\theta^2}{2} + \theta^2$ $= 1 + \theta + \frac{3}{2}\theta^2$ Thus $a=1$ and $b=\frac{3}{2}$	Take note that series contains up to the θ^2 term, hence we are only interested in getting θ^2 from the term $\left(-\theta - \frac{\theta^2}{2} \right)^2$.
5i	Area of $\triangle OAC = \sqrt{126}$ $\Rightarrow \frac{1}{2} \underline{a} \times \underline{c} = \sqrt{126}$ $\Rightarrow \underline{a} \times (\lambda \underline{a} + \mu \underline{b}) = 2\sqrt{126}$ $\Rightarrow \lambda \underline{a} \times \underline{a} + \mu \underline{a} \times \underline{b} = 2\sqrt{126}$ $\Rightarrow \left 0 + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right = 2\sqrt{126}$ $\Rightarrow \mu \left \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right = 2\sqrt{126}$ $\Rightarrow \sqrt{14}\mu = 2\sqrt{126}$ $\Rightarrow \mu = 6$	(i) The most efficient method is to choose vectors $\overrightarrow{OA}$ and $\overrightarrow{OC}$
5ii	$\underline{c} = \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda + 4 \\ -\lambda + 8 \\ \lambda \end{pmatrix}$ $ \underline{c} = 5\sqrt{3}$ $\Rightarrow \sqrt{(\lambda + 4)^2 + (-\lambda + 8)^2 + \lambda^2} = 5\sqrt{3}$ $\Rightarrow 3\lambda^2 - 8\lambda + 80 = 75$ $\Rightarrow 3\lambda^2 - 8\lambda + 5 = 0$ $\Rightarrow (\lambda - 1)(3\lambda - 5) = 0$ $\Rightarrow \lambda = 1 \quad \text{or} \quad \lambda = \frac{5}{3}$ C has coordinates $(5, 7, 1)$ or $\left(\frac{17}{3}, \frac{19}{3}, \frac{5}{3} \right)$.	(ii) Remember to give coordinates for C and not vector $\overrightarrow{OC}$.

6i	$z^3 = (1 + ic)^3$ $= 1 + 3(ic) + 3(ic)^2 + (ic)^3$ $= 1 + 3ic - 3c^2 - ic^3$ $= (1 - 3c^2) + i(3c - c^3)$	Take note of the statement in bold at the start of the question. GC should not be used at all in this question. The $x + iy$ form is required. You can either expand directly or make use of the binomial theorem.						
6ii	$z^3 \text{ is real} \Rightarrow 3c - c^3 = 0$ $\Rightarrow c(3 - c^2) = 0$ $\Rightarrow c = 0 \text{ (NA) or } c = \sqrt{3} \text{ or } c = -\sqrt{3}$ $\therefore z = 1 + i\sqrt{3} \text{ or } z = 1 - i\sqrt{3}$	Since c is a non-zero real number, $c = 0$ has to be rejected.						
6iii	$z = 1 - i\sqrt{3}$ $ z^n > 1000$ $\Rightarrow z ^n > 1000$ $\Rightarrow (2)^n > 1000$ Smallest positive integer $n = 10$ $ z^{10} = z ^{10} = 2^{10} = 1024$ $\arg(1 - i\sqrt{3}) = -\frac{\pi}{3}$ $10\arg z = -\frac{10\pi}{3}$ $\therefore \arg(z^{10}) = -\frac{10\pi}{3} + 4\pi = \frac{2\pi}{3}$ <table><tr><th>n</th><th>2^n</th></tr><tr><td>9</td><td>512</td></tr><tr><td>10</td><td>1024</td></tr></table>	n	2^n	9	512	10	1024	Remember that to get the argument, you should always consider where $z = 1 - i\sqrt{3}$ is on the Argand diagram. There is a need to bring $-\frac{10\pi}{3}$ into the principal range for argument.
n	2^n							
9	512							
10	1024							
7i	Let $y = g(x) = \frac{x+k}{x-1}$ for all $x \in D_g$ $\therefore xy - y = x + k$ $x(y-1) = k + y$ $x = \frac{y+k}{y-1} = g^{-1}(y)$ $\therefore g^{-1}(x) = \frac{x+k}{x-1} = g(x)$ Hence, g is self-inverse.	Note that this is a show question hence it is important to produce all the relevant steps.						
7ii		You may substitute a specific k value in your GC to get the shape, but when you sketch your curve, coordinates should still be in terms of k . Coordinates should be given as instructed by the question.						

7iii	One line of symmetry: $y = x$ ($\because$ self-inverse) $g(x) = \frac{x+k}{x-1} = 1 + \frac{k+1}{x-1}$ $\therefore y = \frac{1}{x}$ Stretch parallel to y-axis by factor $(k+1)$ (x-axis invariant) $\Rightarrow y = \frac{k+1}{x}$ Translate, 1 unit in the positive x-direction $\Rightarrow y = \frac{k+1}{x-1}$ Translate, 1 unit in the positive y-direction $\Rightarrow y = 1 + \frac{k+1}{x-1}$	You need to know the basic connection between a graph and its inverse in order to obtain the line of symmetry. Be precise in the description of the transformation. You should not be using wrong phrases e.g. Shift the curve... Also note that stretch does not require units.
8i	$x - y = (x + y)^2$ $1 - \frac{dy}{dx} = 2(x + y) \left(1 + \frac{dy}{dx} \right)$ $\frac{dy}{dx} (2x + 2y + 1) + 2x + 2y = 1$ $\frac{dy}{dx} (2x + 2y + 1) + 2x + 2y + 1 = 2$ $\frac{dy}{dx} + 1 = \frac{2}{2x + 2y + 1}$	Again a show question, hence all the relevant steps need to be produced.
8ii	$\frac{dy}{dx} + 1 = \frac{2}{2x + 2y + 1}$ $\frac{d^2y}{dx^2} = -2(2x + 2y + 1)^{-2} \left(2 + 2 \frac{dy}{dx} \right)$ $= - \left(\frac{2}{2x + 2y + 1} \right)^2 \left(1 + \frac{dy}{dx} \right)$ $= - \left(1 + \frac{dy}{dx} \right)^2 \left(1 + \frac{dy}{dx} \right)$ $= - \left(1 + \frac{dy}{dx} \right)^3$	
8iii	At turning point, $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} = -(1+0)^3 < 0$ Hence, turning point is a maximum point.	Read the question carefully, and realise that we are interested in the <u>turning point</u> which means $\frac{dy}{dx} = 0$ has to be mentioned.
9i	$\overrightarrow{BA} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} - \begin{pmatrix} -1 \\ -8 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \\ 8 \end{pmatrix}$	Take note of the correct way to write down vector equation of a line Wrong presentation

	$\therefore \vec{r} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$	• Line = $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$ • $AB = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$
9ii	$\overrightarrow{ON} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \text{for some } \lambda$ $\Rightarrow \overrightarrow{CN} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 6+\lambda \\ 2\lambda \\ 6+\lambda \end{pmatrix}$ $\overrightarrow{CN} \perp l \Rightarrow \begin{pmatrix} 6+\lambda \\ 2\lambda \\ 6+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 0$ $\Rightarrow 6 + \lambda + 4\lambda + 6 + \lambda = 0$ $\Rightarrow \lambda = -2$ $\therefore \overrightarrow{ON} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix}$ $\overrightarrow{AN} = \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} - \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \\ -2 \end{pmatrix}$ $\overrightarrow{NB} = \begin{pmatrix} -1 \\ -8 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -6 \\ -12 \\ -6 \end{pmatrix} = 3 \begin{pmatrix} -2 \\ -4 \\ -6 \end{pmatrix} = 3\overrightarrow{AN}$ $\therefore AN : NB = 1 : 3$	Since N is a point on line through A and B, use the equation of line AB to write down $\overrightarrow{ON}$. When 2 vectors are perpendicular to each other, their dot product is 0. Once you know the value of λ, you know $\overrightarrow{ON}$.
9iii	Let C' be the reflection of C in line AB. $\frac{\overrightarrow{OC'} + \overrightarrow{OC}}{2} = \overrightarrow{ON}$ $\overrightarrow{OC'} = 2 \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 0 \\ 11 \end{pmatrix}$ $\overrightarrow{AC'} = \begin{pmatrix} 9 \\ 0 \\ 11 \end{pmatrix} - \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} = \begin{pmatrix} 2 \\ -8 \\ 2 \end{pmatrix}$ Line has equation $\vec{r} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -4 \\ 1 \end{pmatrix}, \quad \mu \in \mathbb{R}$ i.e. $x - 7 = \frac{y - 8}{-4} = z - 9$.	Ratio Theorem is used here. Alternatively, you can consider $\overrightarrow{CN} = \overrightarrow{NC'}.$ Do remember to give cartesian equation for the line (read the question carefully)

10i	Volume, $k = \pi r^2 h + \frac{2}{3} \pi r^3 \Rightarrow h = \frac{k}{\pi r^2} - \frac{2r}{3}$ External surface area, A $= 2\pi r h + \pi r^2 + 2\pi r^2$ $= 2\pi r \left(\frac{k}{\pi r^2} - \frac{2r}{3} \right) + 3\pi r^2$ $= \frac{2k}{r} + \frac{5}{3} \pi r^2$ Let $\frac{dA}{dr} = 0, \Rightarrow \frac{10}{3} \pi r - \frac{2k}{r^2} = 0$ $\Rightarrow \frac{10}{3} \pi r^3 = 2k$ $\Rightarrow r = \sqrt[3]{\frac{3k}{5\pi}}$ $k = \pi r^2 h + \frac{2}{3} \pi r^3 = \pi \left(\sqrt[3]{\frac{3k}{5\pi}} \right)^2 h + \frac{2}{3} \pi \left(\sqrt[3]{\frac{3k}{5\pi}} \right)^3$ $\Rightarrow \pi \left(\sqrt[3]{\frac{3k}{5\pi}} \right)^2 h = \frac{3k}{5}$ $\Rightarrow h = \frac{3k}{5\pi} \left(\sqrt[3]{\frac{5\pi}{3k}} \right)^{\frac{2}{3}} = \sqrt[3]{\frac{3k}{5\pi}}$	A lot of students tend to mis-read the question and hence omit the floor when forming their area expression. h has to be replaced from the volume equation since h is a variable. Always try to tidy up your expression before moving on to differentiate There is no need to prove that area has a minimum value since that is already given in the question. Do remember to find the exact value of h in terms of k as well.
10ii	$k = 200, A = 180$ $\frac{400}{r} + \frac{5}{3} \pi r^2 = 180$ $\Rightarrow \frac{5}{3} \pi r^3 - 180r + 400 = 0$ $\Rightarrow 5\pi r^3 - 540r + 1200 = 0$ This is a cubic equation $\Rightarrow$ at most 3 solutions for r GC gives 2 positive roots and 1 negative roots and since $r > 0$ $\therefore$ there are 2 possible values of r. From the GC, $r = 3.0372$ or $r = 3.72153$ $h = \frac{k}{\pi r^2} - \frac{2r}{3} \Rightarrow h = 4.8765 \text{ or } h = 2.1157$ Since $r < h \Rightarrow r = 3.04, h = 4.88$	There is no need to start again and set up equations for volume and surface area. You should sensibly
11i	$x = \theta - \sin \theta \Rightarrow \frac{dx}{d\theta} = 1 - \cos \theta$ $y = 1 - \cos \theta \Rightarrow \frac{dy}{d\theta} = \sin \theta$	To proceed on after obtaining $\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta},$ you need to realise that 1) trigo functions are involved hence trigo identities may be needed 2) angle needs to be reduced by half

	$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta}$ $= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 - \left(1 - 2 \sin^2 \frac{\theta}{2}\right)}$ $= \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}}$ $= \cot \frac{\theta}{2}$ Gradient of C at $\theta = \pi$ is $\frac{dy}{dx} = \cot \frac{\pi}{2} = 0$ As $\theta \rightarrow 0$, $\cot \frac{\theta}{2} \rightarrow \infty$ Also $x \rightarrow 0$, $y \rightarrow 0$ i.e. tangent is a line parallel to the y-axis that passes through $(0, 0)$ As $\theta \rightarrow 2\pi$, $\cot \frac{\theta}{2} \rightarrow \infty$ Also $x \rightarrow 2\pi$, $y \rightarrow 0$ i.e. tangent is a line parallel to the y-axis that passes through $(2\pi, 0)$	Note that trigo ratio of all special angles like $\frac{\pi}{2}$ has to be evaluated. Note that the question is asking about tangents (which are straight lines), NOT gradients, hence your answer has to mention the specific tangents whose gradient $\rightarrow \infty$ for the respective θ values.
11ii	(ii) 	Remember to label the coordinates of the maximum point and axial intercepts. You must show clearly the shape of the curve at both $\theta = 0$ and $\theta = 2\pi$. Use your answers from part (i).
11iii	$\text{Area} = \int_0^{2\pi} y \, dx$ $= \int_0^{2\pi} (1 - \cos \theta)(1 - \cos \theta) \, d\theta$ $= \int_0^{2\pi} (1 - 2 \cos \theta + \cos^2 \theta) \, d\theta$ $= \int_0^{2\pi} \left(1 - 2 \cos \theta + \frac{\cos 2\theta + 1}{2}\right) \, d\theta$ $= \left[\theta - 2 \sin \theta + \frac{\sin 2\theta}{4} + \frac{\theta}{2}\right]_0^{2\pi}$ $= 3\pi$	Learn how to find area under a curve that is defined parametrically without finding the Cartesian equation. Note that the limits for x and for θ happen to be the same values. Again, use your working from part (i) to see that.

11iv	$\frac{dy}{dx} = \cot \frac{p}{2}$ $\Rightarrow \text{gradient of normal at } P = -\tan \frac{p}{2}$ Equation of normal at $P(p - \sin p, 1 - \cos p)$ is $y - (1 - \cos p) = \left(-\tan \frac{p}{2}\right)(x - p + \sin p)$ When $y = 0$, $0 - (1 - \cos p) = -\left(\tan \frac{p}{2}\right)x + \left(\tan \frac{p}{2}\right)p - \left(\tan \frac{p}{2}\right)\left(2 \sin \frac{p}{2} \cos \frac{p}{2}\right)$ $-1 + \left(1 - 2 \sin^2 \frac{p}{2}\right) = -\left(\tan \frac{p}{2}\right)x + \left(\tan \frac{p}{2}\right)p - 2 \sin^2 \frac{p}{2}$ $\left(\tan \frac{p}{2}\right)x = \left(\tan \frac{p}{2}\right)p - 2 \sin^2 \frac{p}{2} + 2 \sin^2 \frac{p}{2}$ $\left(\tan \frac{p}{2}\right)x = \left(\tan \frac{p}{2}\right)p$ $x = p$ Hence normal to C at P crosses the x-axis at the point $(p, 0)$.	This is a difficult part and a lot of errors were made. Read the question with care and take note of the following 1) We are looking for the normal at $P(p - \sin p, 1 - \cos p)$ NOT any other point. 2) Try to work backwards from the given answer i.e. $x = p$ (answer) $\Rightarrow kx = kp$ Hence all the rest of the terms in the equation of normal should eventually be cancelled off.
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